A review of non-invasive BCI devices

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Abstract: BCI provisions humans beings to control various devices with the help of brain waves. It is quite useful for the people who are totally paralysed from neuromuscular diseases such as spinal cord injury, brain stem stoke. BCI permits a muscular free channel for conveying the user intent into action which helps the people with motor disabilities to control their surroundings. Various non-invasive technologies like electroencephalogram (EEG), magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI), etc. are available for capturing the brain signal. In this article, various non-invasive BCI devices are analysed and nature of signals captured by it are reported. We also explore the use of signals for diseases diagnosis, their features and availability of those devices in the market.

Keywords: electroencephalogram; EEG; magnetoencephalography; MEG; functional magnetic resonance imaging; fMRI; non-invasive; psychological; physiological diseases.

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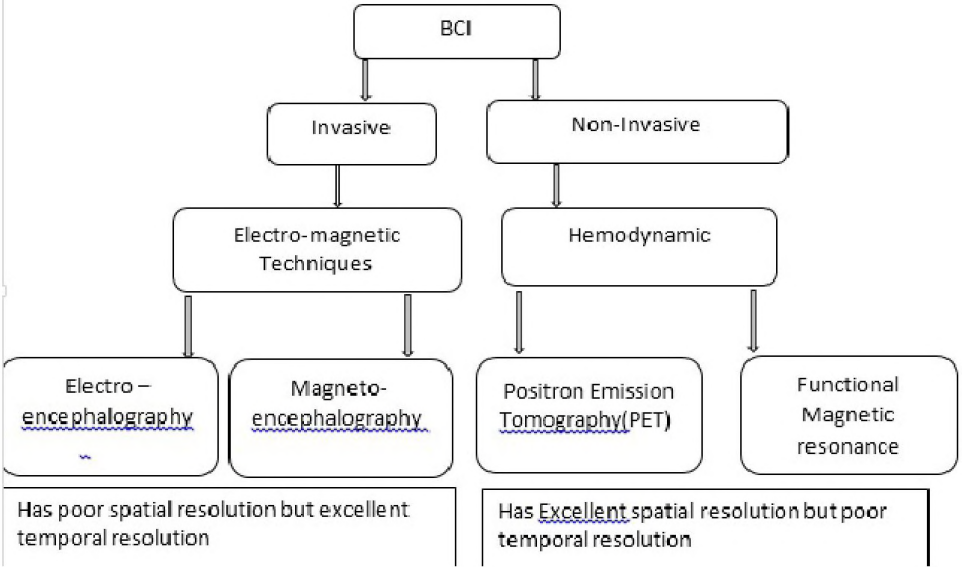
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1 Introduction

The soul of India lives in its villages, 60% of the population still lives in villages of India. The educational status and facilities available in villages, particularly medical facilities are very poor. The hospitals in rural areas suffer from non-availability of doctors, equipments and technicians, so the village people have to reach nearby cities to get better health care. Many times the diseases reach chronic stage by the time patients reach city hospitals and have coasted to loss of many lives. There is need for early diagnosis of diseases and initiation of immediate treatment for the patients residing in rural areas which can cure many diseases at early stages, and will stop spreading epidemic diseases. In the olden days due to lack of technology they were not able to cure the diseases in rural areas.

Today we have technology that supports to cure the diseases remotely using the various neuroimaging methods. Such as invasive and non-invasive methods as in Figure 1.

Figure 1 Non-invasive functional brain imaging techniques



1.1 Invasive

Generally invasive BCI uses a set of sensors (electrodes) that will be placed into grey material of the brain through neurosurgery. These invasive BCI produce good quality output as the electrodes are inserted into the grey matter of the human brain. A drawback of invasive-BCI is that the electrode signal become weaker due to build-up of scar tissue also the human brain reacts to the foreign objects (electrodes).

1.2 Non-invasive

In non-invasive BCI methods has less signal clarity as the devices are placed on the surface and they are harmless. Figure 1 shows non-invasive functional brain imaging Techniques. In this survey, emphasis is mainly on non-intrusive methods. The main motivation for this work is to check the feasibility for easy diagnosis of physiological diseases in humans. Table 1 lists these different non-invasive devices.

Table 1 Non-invasive BCI acquisition methods

Sl no.	*Non-invasive method*	*Description*
1	Electroencephalogram (EEG)	Electrical signals are recorded using flatmetal discs
2	Magnetoencephalography (MEG)	Magnetic fields are recorded using neuroimaging method
3	functional magnetic resonance imaging (fMRI)	Measures the variation of blood oxygen level during brain activities
4	Positron emission tomography (PET)	Imaging technique used to observe the metabolic process in body
5	Near-infrared spectroscopy (fNIRS)	Near-infrared spectroscopy uses electromagnetic spectrum to diagnose physiological diseases
6	Fetal magneto encephalography (fMEG)	Study of brain activity of fetus in uterus
7	Single photon emission computed tomography (SPECT)	Neuro tomographic imaging which is based on gamma rays

Even though we have various non-invasive devices, each of the devices has its own advantages and disadvantages with respect to various parameters. In order to help the remote diagnoses where expert doctor is not available the device that can be suggested is EEG, with the availability of a nurse in the clinic the device can be used to collect the signals. EEG devices has the following advantages like cost of the device is less, its ease of use, portable, safer and also helps to diagnose many psychological disease. Using this device we can also predict the physiological disease by capturing the brain signals and these signals can be sent to the expert doctor present in urban cities for diagnosis. There are various types of EEG devices available in the market which varies in number of electrodes, cost, purpose, etc. so EEG can be used to predict the psychological disease.

The sub-sequent sub-section brief about each device, its modes, operations and capabilities.

2 Literature survey

Pedreira et al. (2009) has discussed inspiration for BCI systems came into existence for aiding the physically challenged people, by capturing the motor signals from brain and using it to move their body parts. Many people who are incapacitate or have undergone amputation such people may see the object what they want but they cannot reach that object due to disability, for example they see a glass of water, and want to reach it. Through the use of BCI they can process certain commands in their brain to achieve their

goal using neural prosthetic device like robot arm. This device can read the brain signals of the people, and perform the required activity.

The brain activity is measured using BCI which represent the user intention, and to convert the logged brain activity into equivalent control signs for BCI applications. The pattern recognition techniques and signal processing techniques are employed to convert brain signals to actions. Graimann et al. (2009) referred the system as brain computer interface, because the activity is directly measured from the brain and not from the muscles or other parts of body. BCI is also called as human computer interface or brain system interface.

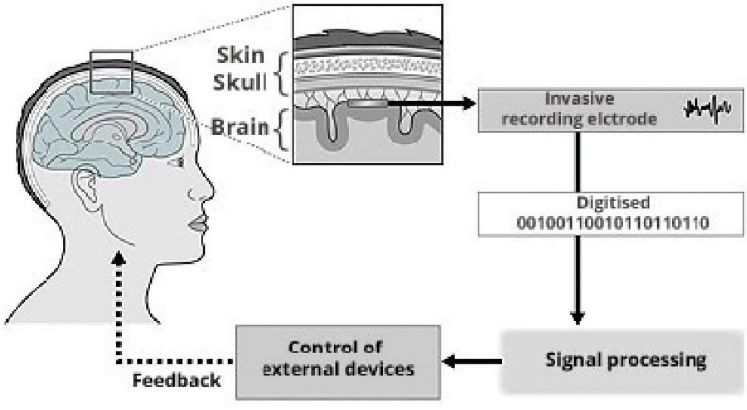
Dr. Grey Walter in 1964 initiated the preliminary work on BCI. In his study, he recorded the patient's brain activity by placing the electrodes on the brain directly, when the operation was in progress. The objective was to capture the patients movement, whenever a task such as pushing a button (imaginary thought by the patient to slide the projector was initiated. This was noticed, when the projector advanced because of the task performed by the patient.

2.1 Electroencephalogram

2.1.1 Electroencephalogram signals

Nicolas and Gomez-Gil (2012) showed how electroencephalogram (EEG) signals are captured by placing the electrodes around the head. It is most commonly used non-invasive method. The EEG signals are weak compared to invasive-signals as the signals have to pass through several layers of the head as in Figure 2.

Figure 2 Non-invasive BCI (see online version for colours)

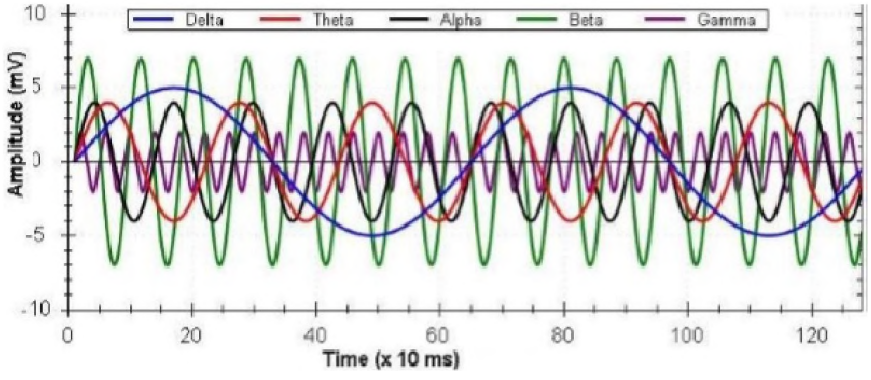


It has various frequency bands as shown in the Figure 3. The EEG signal contains a lot of noise introduced due to blink of an eye, muscles movement, etc. The placement of electrodes on the scalp is according to International 10–20 system as in Figure 4.

There are five frequency bands in EEG signals they are as follows: Gamma, Beta, Alpha, Theta and Delta. Figure 3 shows all the five frequency bands.

- *Gamma:* The Gamma frequency range is 31 Hz and above. It reacts the awareness of human brain. Gamma and Beta waves together represent the extent of attention, perception, and cognition of human brain.
- *Beta:* The Beta waves frequency range between 12 to 30 Hz, They are further classified into Beta1 and Beta2 to obtain required choice. The beta-waves are fast and minute, represents attentive awareness. These waves are found in central and frontal areas of brain. The beta activity will increase during situations such as solving a math problem or while resisting or in suppressed movement.
- *Alpha:* The frequency range of Alpha-waves is between 7.5–12 Hz, they represent relaxation and disengagement. The alpha activity increases while closing the eyes and thinking with peace. The alpha signals are found in back of the head and in the forward lobe.
- *Theta:* The Theta waves frequency range between 3.5 to 7.5 Hz, these waves represent inefficiency, carelessness, and the lowest waves of theta represent whether a person is awake or in sleep state. Theta activity increase during emotional stress, frustration or disappointment. Theta waves also represent unconsciousness distracted, unfocused and deep meditation states of human activity. Increased altitudes of theta waves indicate symptoms of attention deficit hyperactivity disorder (AD/HD) in adults.
- *Delta:* The Delta waves frequency range between 0.5 to 3.5 Hz, these waves have highest amplitude and slowest waves which represent deep sleep. Delta waves happen to be in different regions of the brain in different extent. If the delta waves are detected in awake state, it determines defects in the brain. Observing raw EEG records can detect the artificial delta waves that occur due to movement.

Figure 3 The five frequency bands and their relationship (see online version for colours)



The primary use of EEG is to study the brain disorders like epilepsy. A brain is considered to be in dead state if it is not generating any signals when observed through EEG. The brain functional map as in Figure 6 can be obtained by using EEG is as shown.

Figure 4 Electrode over scalp in 10/20 system, side view

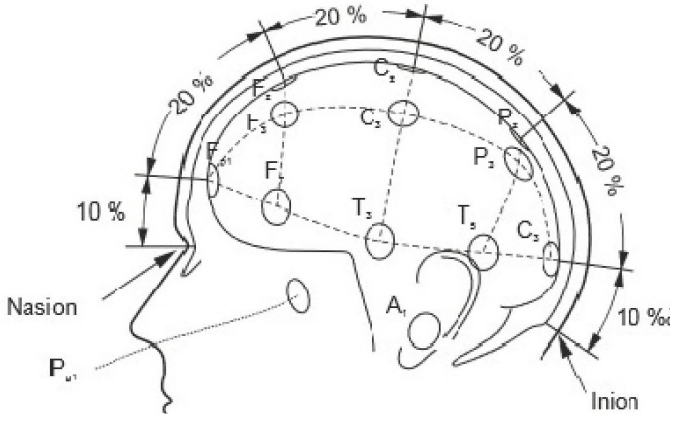
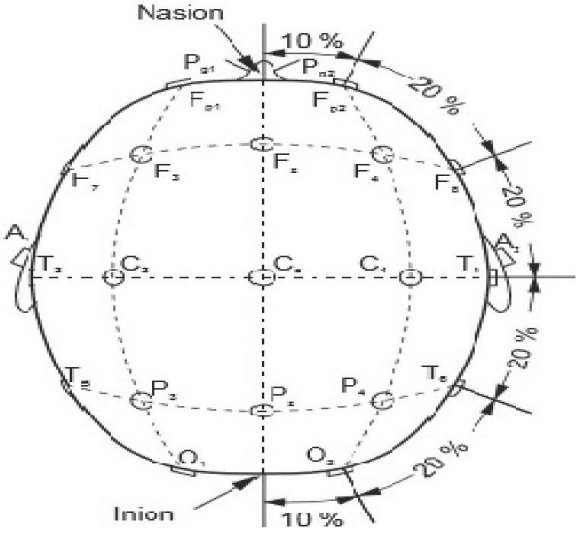


Figure 5 Electrode over scalp in 10/20 system, top view



When the subjects brain is exposed to external stimulus, such as a tone or flash light, the brain generates EEG waves in response as shown in Figure 7.

Graimann et al. (2009) discussed that when the subjects brain is exposed to internal stimuli, such as skipping will also generate EEG signals in response. The above two situations are known as event related potentials (ERP), or evoked potential (EP). ERP/EP indicate the existence of a visible amplitude crest (potential) that happens by a specific time delay after each stimuli. The commonly used ERP is P300 (P3), in which the peak of the wave occurs 300 milliseconds post event occurrences. In medical diagnostic P300 application is used to relate the brain activity to external stimuli in particular regions of brain. For instance, if the brain is exposed to external stimuli like a tone, then the EEG waves will be generated due to reaction close to the hearing dispensation area. The comparision of steady state visually evoked potential (SSVEP), P300 and slow cortical potentials (SCP) is shown in Table 2.

Figure 6 Basic functional brain map (see online version for colours)

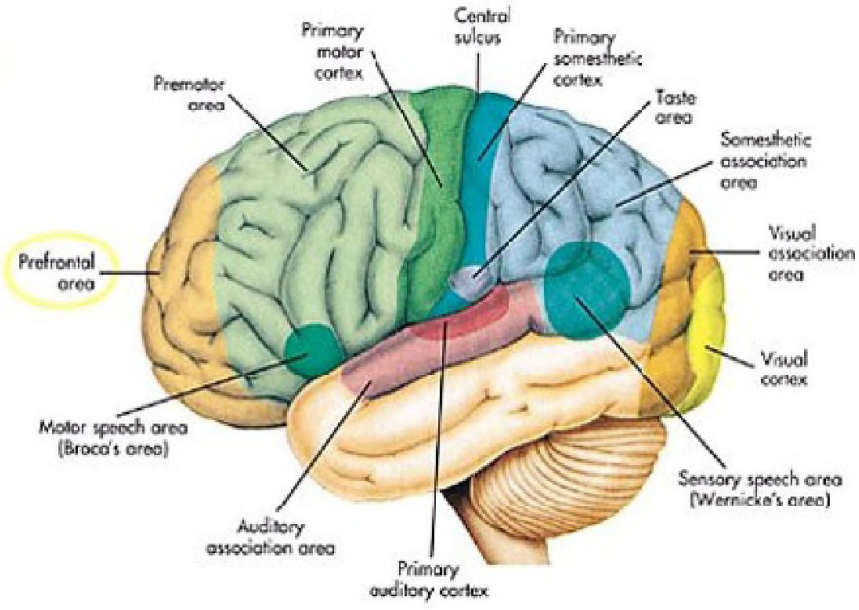
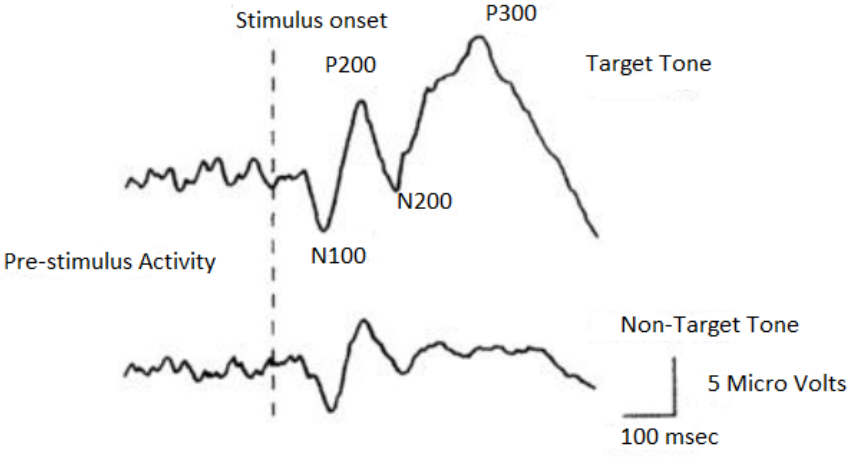


Figure 7 Sample evoked responses in brain



2.1.2 Types of EEG

1 *SSVEP:* Palaniappan (2014) has discussed how SSVEP brain signals produced due to visual stimulation. When the subject views flashing light at a specific frequency, the flashing lights stimulate the visual pathway; this stimulation produces electrical signals of varying frequencies. When the visual stimulus is flickered above 6 Hz frequency then SSVEP occurs. The subject observes the flashes in a pattern and flashing light flickers in a particular frequency on screen. The light flashes create

EEG which waves in the recurrence as that of the flickering object. This effect is called as photic response of the brain. These spontaneous responses do not involve manual effort, except that the subject needs to gaze at the stimulus.

2 *P300 visual evoked potential (VEP):* Palaniappan (2014) has discussed about P300 VEP is an EEG system developed for visual stimulus. The frequency of VEP will be around 300,600 ms and is high in midline locations such as Fz, Cz and Pz. VEP signals are filtered using low pass filters before analysis as their frequency is limited to 8 Hz. VEP is used in variety of decision-making tasks like identifying target stimulus for example when a picture is to be recognised.

3 *SCP:* Palaniappan (2014) has discussed an SCP are slow current shifts in EEG based on the events. The SCP signals can occur between 300 ms to 1 s. It measures the electrical activity in brain produced in response of a stimuli. Various tasks can be controlled by using positivity or negativity of SCP. Positive SCP is generated during cognitive tasks or relaxed states. Readiness/planning activities can generate negativity SCP. To obtain high performance they need rigorous training for SCP so it is not a popular EEG device.

Table 2 Advantages and disadvantages of SSVEP, P300 and SCP

Parameters	*SSVEP*	*P300*	*SCP*
Characteristics	Based on signal modulation in visual cortex	Positive peaks due to infrequent stimulus	Based on modulations synchronised to motor activities
Transfer rate/bits	60–100	20–25	5–12
Training	No	No	Yes
Advantages	Needs less training and high bit rate	Single EEG channel is required	Does not depend on any stimulation
Disadvantages	Permanent attention to external stimuli	Some subject may get tired	Needs more amount of training time

2.1.3 Disease diagnose with EEG

Larsen (2011) has discussed about EEG as commonly used to diagnose AD/HD. Individuals with AD/HD as a rule will have moderate waves, thus when these individuals concentration is in forehead or frontal lobe it gets difficult for them to oversee consideration, conduct and control motivation.

Smith (2005) discussed about a patient whose EEG indicates triphasic waves and diffuse moderate action with intense change in mindfulness will be inclined to metabolic encephalopathy. In a patient of uraemic encephalopathy having acute deterioration of renal function, EEG changes are more severe. The association between extent reduction of EEG signal and relative hike in serum creatinine can be observed. Spike wave epilating from seizures may happen in any serious encephalopathy. EEG is used to detect the slow activity of focal/lateralised rhythms and focal neurological deficits in hypoglycemia patients. EEG can be helpful for early identification of dialysis dementia that occurs in some long-term dialysis patients caused due to aluminium intoxication as

the clinical symptoms are seen only after several months in this disorder EEG is additionally utilised as a part of conclusion of endocrine diseases. EEG changes are similar with the seriousness of hormonal unsettling influence, the separation between essential endocrine and auxiliary electrolyte impact is troublesome. Hashimoto's thyroiditis has varied clinical and electrographic diagnosis; EEG is used to diagnose Hashimotos thyroiditis which shows lateralised temporal slowing, frontal slow activity, periodic sharp waves and triphasic waves. EEG is used to determine neurodegenerative complication through pathognomonic variations caused due to measles virus infection in childhood. The HIV encephalopathy happens due to damage to the brain by longstanding HIV leads to problems of concentration, memory and slowness of physical movements. In such cases mild EEG slowing is observed, also epileptiform discharges occur in association with seizures. The observations also show prominent intermittent temporal slow activity.

Here the author (Singh and Bansal, 2016) has used the EEG EMOTIV head set which consists of 14 channels to collect the data of different samples from different age groups they have used a MATLAB platform by developing an custom-based application program interface (API). Here the author (Sabeti et al., 2016) has discussed how to classify the patients with schizopheria from control subjects using EEG signal data.

Here mainly the author (Singla and Jatana, 2017) has highlighted that cost of EEG Devices are less, safer, ease of use. They have highlighted on the drawbacks of traditional methods of EEG and tried with the hybrid methods of the BCI and have discussed about different applications. In this paper, Pritika and Budhiraja (2016) they have discussed how to merge more than one image modalities so as to extract the overall information which helps the radiologist for easy and fast diagnosis of the disease as each modalities highlights on dissimilar information here they have shown that traditional methods has some drawbacks they have used new methods.

Here, Boukari and Djemili (2016) have discussed classification of EEG signals using neural networks to observe how EEG signals will vary from diseased person (seizure) to the normal person. Even for the trained neurologist it is difficult to predict the disease using EEG signals has the amplitude of signals will be less and contains lot of disturbances so in this paper they have tried to automate the seizure detection which will be useful for diagnosing the diseases. Akareddy and Kulkarni (2013) discussed about detecting an epilepsy seizure using the hybrid method of EEG signals which is a combination of multi wavelet transform and artificial neural network.

Here, Kidwai and Saeed (2017) discussed about the percentage of anaesthesia dose that has to be given to a patient, because what amount of anaesthesia should be given is not known given in excess may affect the human health so they use EEG signals to monitor. Here, Ramadan and Vasilakos (2017) have discussed how BCI helps the physically disabled people to live like normal persons and brief review on the BCI hardware and software technologies that is used, BCI control signals, its types and classifications.

2.1.4 EEG devices in market

There exists a variety of EEG-based BCI devices in market manufactured by different companies. Below is a list of such EEG-based BCI devices

2.1.4.1 Emotiv

The software provided with Emotiv is based on the machine learning algorithm which is capable of learning a phenomenon by human perception or based on how humans visualises it, for instance movement of objects, from the EEG measurements. These EEG measurements are mapped to computer controls. The neuroheadset of Emotiv consists of 14 different sensors, includes a gyroscope. The device costs around 300 USD.

2.1.4.2 ULandT

ULandT is utilised in home training using computer with special programs and games for subjects with AD/HD or attention deficiency disorders. The ULandT adopts neurofeedback technology. This system is not only used in household environment, but also used by schools, hospitals and psychologists.

2.1.4.3 SmartBrain

It uses neurofeedback with play stations and Xbox and has been applied to thousands of games. The system has inbuilt controller when the user concentration is high, the controller becomes more responsive. There are two commercial version of SmartBrain:

1 Personal edition costs 600 US dollars and available with three training presets.

2 Professionally supervised costs 800 US dollars provides options to adjust the individual parameters based on user preferences. Smart Brain increase beta brain waves and reduce alpha and theta mind waves.

2.1.4.4 NeuroSky

The devices include Mind Wave containing ten applications including games educational applications for children to learn mathematics, memory tests and pattern recognition. The force trainer uses the power of mind to move multi-dimensional images and perform astounding deeds of Jedi. The device calculates the attention and meditation level. The object will be lifted if the meditation level increases; the stuff gets burnt if the attention level increases. The output of the Neurosky headset is EEG signals.

2.1.4.5 Mind ball

The user wears a headband that reads the brain waves from the forehead. Alpha waves (drowsiness) and theta waves (relaxation) are used to manage the movement of a ball. These signals move the ball from one point to other in a game with a challenger. The user who is having more concentration pushes the ball towards the opposite end, and wins.

2.1.4.6 Olimex EEG-SMT

Is an open source low cost EEG device, which uses USB interface. The prime use of the device is to assess biofeedback from various sources apart from the brain. Forthcoming events will consist of finding the modifications in patterns of electrical action of the brain as an alternate method to transcranial direct current stimulation (TDCS). Other additional projects contain examining biofeedback and transcranial magnetic inspiration.

2.1.4.7 SIENNA ULTIMATE

It is a hybrid (long-term monitoring) LTM and ambulatory EEG amplifier developed by EMS biomedicals. It has 32 unipolar or bipolar channels, 32 GByte memory card, incorporated Lithium-ion battery, SaO_2 and WIFI interface. The Intercom feature provides direct communication means between a patient in a lab and nurse station or in a residence care. The Real-time battery charge information is sent to nurse to conserve it. The real-time EEG display is available at the nurse station, even though the patient is out of the building. The SIENNA ULTIMATE can be adapted to use with head boxes for cerebral function monitoring (CFM), LTM amplifier, sleep, 10/20 and 32 channel bipolar EMG for sports medicine. The body position sensor and patient alarm button is placed in a 150 gram lightweight amplifier, which transfer fall over information to the nurse station. The SIENNA ULTIMATE can do Ambulatory recording for a maximum of 4 days.

2.1.4.8 VIRGO-24/32/40

Is a light weight EEG device manufactured by Allegers. It is easily portable and can be used in ICU/bed side. It comprises of USB 2.0 interface to transfer the data, high-resolution MPEG video and an integrated impedance check for electrode connectivity. This unit functions at 230 volts and 5 ampere supply.

2.1.4.9 Be micro

Is crafted by EBNeuro, is a smallest device for EEG monitoring. This compressed device offers both ambulatory EEG monitoring and traditional EEGs. It also checks the oxygen threshold and electrocardiogram simultaneously while recording the electroencephalography. This device can be combined with most of the clinical environments with a USB cable. The Be Micro is friendly with most EEG electrodes. The device has an LCD screen to experiment achievement parameters, check impedance stages, and perceive EEG signals. The functions of Be Micro are user-friendly application process. It also comes with multiple task joysticks, which allows patients to access incident histories.

2.1.4.10 Brainnet4

The MEDATECs 24 bit direct digital Brainnet4 EEG offers continuous non-intrusive EEG monitoring, lively reference transferring, etc. Brainnet4 is the faultless application for excellent EEG recordings in any situation such as routine, long-term checking, day or night epilepsy diagnosis, ICU, Neonatology, operating room, etc.

2.1.4.11 Brainfeedback pro

Deymeds BrainFeedback Pro contains dedicated neurofeedback, 3D games, Z-score training, trend graph reports and an advanced protocol setup utility. It has six customisable threshold bins. It is the most user-friendly neurofeedback system.

2.1.4.12 Neuro prax

The Neuroprax EEG-9100 of NIHON KOHDEN is a notebook PC-based EEG, works on Windows XP. The performance and scalability is high. The system is provided with uncompelled software and hardware. The EEG testing can be accomplished at any place using Neurofax. It is a ready to use complete system furnished with a separate power supply, Multi-function electrode junction box, a notebook PC, accessories and standard software. It is frivolous and handy, so consumers can smoothly transfer the system to a car or airplane. The quality of EEG recordings can be guaranteed, even in noisy ICU or ER. The Neurofax is also provided with spike detector software and analysis software for off line spike detection.

2.1.4.13 Schwarzer LTM

Developed by NATUS, is a long term monitoring device. It incorporates Deltamed Coherence Software which deals with several functions in an elastic capacity and has various time-saving characteristics. It enables EEG and polygraph recordings with 40, 64, 128, 192 or 256 channels. It supports Full-HD Video recording using network cameras based on H264 compression. Schwarzer LTM provisions preview and control of videos from various rooms on a single monitor. It provides automatic switching of cameras between the rooms. It has smallest LTM amplifiers (240 gr/64 channels). It has Daisy chaining 64-channel amplifiers. It provides different analysis functions such as EEG analysis seizure, spike. Fileviewer provides summary of several days recorded data.

2.1.4.14 ATES neurotravel MINI

Is an 8 channel PC-based EEG device provided with USB interface. It can be used for monitoring EEG in intensive care unit (ICU) and emergency/medical tests. Neuro Prax is a direct current (DC) EEG system used to measure the physiolgical signals such as EP, EEG and EMG whose frequency ranges between 0–1200Hz. Neuro Prax amplifier can measure slow to ultrafast EEG activities. It is a table mode device.

Table 3 shows the comparision of different non-invasive devices available in the market.

2.2 Magnetoencephalography (MEG)

2.2.1 MEG signal

Sheridan et al. (2010) discussed that when the neurons in the brain are triggered, electric current flows which in turn creates magnetic field. These actions are called neuroelectromagnetism. In MEG, these magnetic fields are mapped to brain activity and are represented in a graphical form. MEG is non-invasive technique that has excellent spatial resolution and time resolution as in Figure 8 the signal analysis for MEG is shown in Figure 9.

Here the authors (Chowdhury et al., 2013) have discussed how magnetic signals from the brain to used to detect epilepsy as it over performs EEG signals.

Table 3 Different non-invasive EEG devices available in the market with capability and applications

Devices	*Capability*	*Application*
Mindflex	It uses 1/3 electrode(s). Measures theta wave activity	Deep and raw emotions can be captured
Mindball	It uses 1/2 electrodes in each headband. Measures alpha and theta waves	Can be used to check if patient is calm or agitated in mind
Force trainer	The electrodes are dry neural sensors. It uses 1/2 electrode(s) in each headband. Measures beta waves	Help to translate beta waves to physical action
NATUSs Schnauzer EEG	It is able to facilitate EEG and polygraphy recordings with either 29 or 40 channels	More suitable for hospital environment for, monitoring patient health
ATES Neurotravel MINI	8 channel EEG device	More suitable in clinical environment for patient monitoring
Neuro Prax EEG	EP, EEG and EMG signals of frequency 0–1,200Hz are measured. Neuro Prax amplifier can measure slow to ultrafast EEG activities	Can be used only in clinical environment
Be Micro	It also monitors oxygen saturation and electrocardiogram at the same time using electroencephalographic recording technologies	Can be used for patient health monitoring
Olimex EEG-SMT	Measure biofeedback from brain and other parts	Can be used to measure brain response to drugs

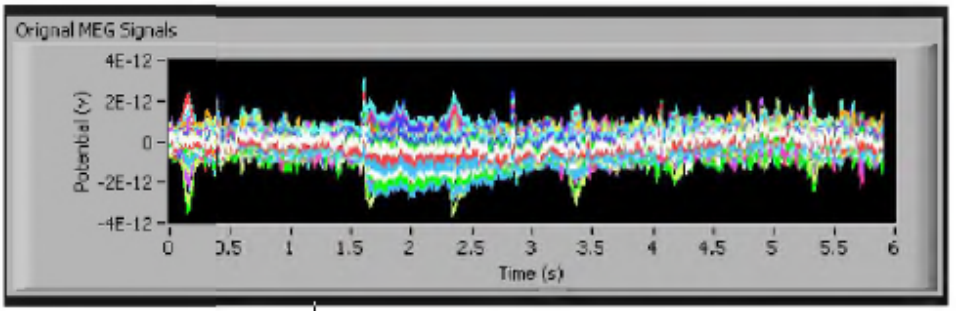
Cohen (1968) has conducted the primitive bio magnetic measurements called Magnetocardiographic (MCG). For the study they used two comparable spirals mounted on a ferrite core, each with a substantial number of windings. The spirals were wired in sequences with contradictory divergence, to work as a gradiometer which can measure the neighbouring bio magnetic field filtering the surrounding noise fields. This method was effective, but resulted noisy measurements then onwards, Cohen (1968) started making comparable measurements, using a low noise amplifier and measurements were conducted in magnetically shielded room, which produced better output. Cohen used this system to conduct primitive MEG measurements in 1968, but resulted in poor signal to noise ratio. Later during 1969, Jim Zimmerman invented superconducting quantum interference device (SQUID), the device was capable of detecting mild magnetic signals. Later, Cohen (1972) and Zimmerman et al. (1970) associated together for bio-magnetism research and they developed the MCG device which can generate noise less MCG signals. Cohen (1972) discussed the commercial production of the SQUID devices started in 1971 which was used by Cohen for first MEG recording. Present MEG devices made up of hundreds of magnetometers, are capable of measuring neuromagnetic fields. The whole-head MEG devices comprise of a large dome like structure hollow cavity where the subjects head is placed while scanning. The SQUID sensors are placed in the inner part of the dome which measures the magnetic signals from brain.

Figure 8 The MEG system



Source: Sheridan et al. (2010)

Figure 9 ICA for MEG signals analysis (see online version for colours)



2.2.2 Disease diagnose with MEG

Hatsusaka et al. (2009) proposed the use of MEG for diagnoses of Alzheimer disease. The subjects were made to hear small beep sounds with a periodic pause. Arbitrarily few beep sounds were removed from the sequence and this elimination of beep sounds is known as tone omitted event. The researchers have emphasised on the explicit magnetic field element produced due to tone omitted event. The researchers have chosen 32 subjects with early Alzheimer's disease (AD) and 32 normal subjects. Both the group of subjects were scanned using 160-channel MEG device and the signals were recorded individually for tone omitted event. The observation showed that the amplitude of the average MEG signals was low in t he AD group compared to the normal group of subjects. Hence this demonstrates the early diagnosis of AD using MEG scans.

James et al. (2015) discussed about post-traumatic stress disorder (PTSD) as a mental disorder seen in people who have witnessed scary, disastrous situations in life. The major characteristics of the disease are flashbacks, hallucination and severe anxiety. James et al. (2015) had used MEG for diagnose and treat patients of PTSD. Chu et al. (2015) studied 250 subjects for both normal and affected PTSD. The MEG scans are conducted placing the subject under the dome of MEG device. The patient needs to concentrate on the light spot for a minute. The scans can reveal the unstimulated idle state of the brain and brain map. The MEG patterns show the difference in neural communications between healthy participants and people with PTSD disorder. Aphasia is a mental disorder related to communication caused due to injury to the left hemisphere of the brain that controls

language. Aphasia causes complications in listening, speech, writing and reading. The main cause for this disease is stroke. Chu had used MEG for diagnosis of strokes. Ogawa et al. (1993) has discussed about multiple sclerosis (MS) is a demyelinating disease where the protecting covers of brain nerve cells are damaged. The symptoms are partial blindness, double vision, weakness. The nerve conduction will be slower in MS patients. Ogawa et al. (1993) used MEG for early diagnosis of MS disease.

2.2.3 MEG devices in market

SQUID is a highly sensitive magnetometer that can detect measure mild magnetic signals. MEG devises adopt DC SQUIDs, made up of two parallel Josephson junctions in a superconducting ring. This superconducting ring is either niobium or alloy of lead, indium and gold. To constantly maintain superconducting property the device needs to work in absolute zero temperature. The electrons moves in both the directions of the ring concurrently. The external magnetic field induces the current in the SQUID sensors. These SQUID sensors are connected to signal cable. The changing amplitude of the SQUID produces the MEG signals.

Vector view is a commercial MEG system developed by MINDSET. It contains 306 SQUID detectors surrounding the head, 102 magnetometers and 204 planar gradiometers. EEG data is captured up to 5 kHz sampling frequency. The system is placed in a huge fluid helium inside the white gantry. It contains 306 pickup coils internally; three different coils form one detector these detectors are placed at 102 different locations on the head. The system costs around 3 million US dollars.

Elekta Neuromag is a cutting-edge MEG device. It can measure magnetic fields generated by the brain non-invasively and maps it to the real-time brain activity. Elekta Neuromas exclusive sensor design and software enables it to collect data with finest details from the deepest parts of the brain. The system has the highest resistance to magnetic interference, either patient related or external. Elekta Neuromag also has internal helium recycler, hence leads to lowest operational costs.

2.3 Functional magnetic resonance imaging (fMRI)

2.3.1 fMRI signal

fMRI is a kind of magnetic resonance imaging (MRI), usually used to deliver structural pictures of human body. In the specialised application of MRI, it is used to produce tomographic images of variations in hemodynamic responses which represent the temporal changes in neural activity. In fMRI, a high Tesla magnetic field combined with radiofrequency (RF) pulse and magnetic gradient is used to image the patient's brain, based on the principle of nuclear magnetic resonance (NMR). It is neither radioactive nor invasive with few known side effects.

Ogawa et al. (1993) discussed the initial breakthrough for fMRI came from Seiji Ogawa when he investigated the RF signals when the brain functions. He worked with the knowledge that activated brain cells used more oxygen than cells at rest. A deoxy-haemoglobin is paramagnetic and changes the magnetic field around it. This distortion of the magnetic field, in turn, affects the magnetic resonance of nearby water protons amplifying their signal as much as 10,000 times. Ogawa called this effect blood oxygenation level dependent (BOLD) contrast imaging.

Logothetis (2002) discussed fMRI usually measures the BOLD signal on a voxel by voxel basis, which increases with high brain activity. The continuous functional scan can monitor a patient's health continuously but is not feasible due to the cost of the equipment and shielding requirements. It is also technically impossible, as the BOLD signal will saturate under long exposure of a constant strong magnetic field. Patients installed ferromagnetic with metallic implants and Cardiac pacemakers are affected by the strong magnetic field. Hence they cannot be scanned using MRI machine. Shellock and Kanal (1996) discussed the switches inside a cardiac pacemaker could become inoperative under a static magnetic field of 0.2 mT. Ogawa et al. (1993) discussed an Ferromagnetic metallic implants such as surgical clip, could be twisted due to influence of the static magnetic field this twisting effect could cut vital blood vessels. The strong magnetic field may also cause eddy currents in metallic implants and devices, which may heat up and might cause damage to the surrounding tissue. Non-ferromagnetic metal implants can lead to obstacles in scan, also if the implant is near the region of interest. Pregnant women are advised against undergoing MRI scans, as the strong magnetic field can cause adverse effect on fetus.

Here the authors (Sato et al., 2013) have used the functional MRI neuro feedback which gives more spatial resolution and since it has gauge deep cortical and subcortical brain regions against EEG neuro feedback signals.

2.3.2 Disease diagnose with fMRI

Huntington's disease (HD) is a neurodegenerative disorder caused due to autosomal dominant mutation of huntingtin gene which provides the genetic information for huntingtin protine. The development of cytosine adenine guanine (CAG) triplet rehashes in the gene coding for the Huntingtin protein results in an anomalous protein, which harms the cells in the brain bit by bit. The HD is portrayed by degeneration in the caudate and putamen neuron loss in cortical layers 3 to 6. The symptoms of HD appear in middle age which include mood swings, thought problems, frequent jerks during body movements, reduced reasoning and thinking abilities also may lead to heart diseases. The clinical diagnosis for HD is usually conducted on finding the motor problems but the neuropsychological deficiencies, structural and functional changes in brain and the psychiatric symptoms will appear much earlier to motor symptoms. fMRI can be used to study basal ganglia function for early diagnosis of HD in presymptomatic individuals.

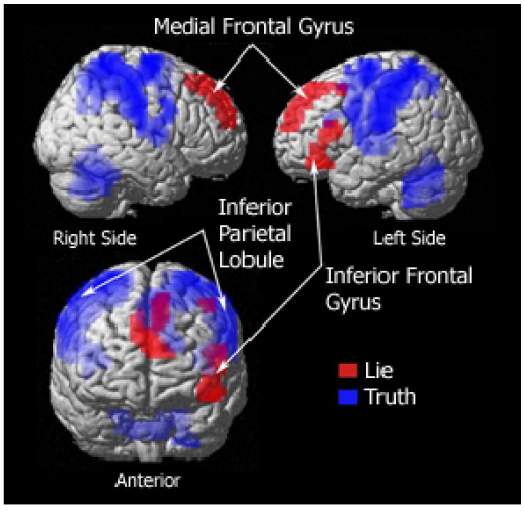
Schizophrenia is a mental disorder in which subject will lose the ability to recognise what is real and have abnormal social behaviour. The patients may encounter hallucinations like listening to voices or daydreams and confused thinking and speech. The disorganised thinking may result in loss of thought, speaking incomprehensible words, no concern about hygiene and dressing, etc. fMRI scans are used in effective diagnosis of schizophrenia.

2.3.3 fMRI devices in market

Nata Technologies develops customised fibre optic sensing products for functional MRI and MEG scanners such as fMRI mouse, joystick, button pad, numerical pad and vibrator system. NeuroDevice is a Poland product-based company developed as an open source software for fMRI reporting referred as brain activation localiser. It can automatically produce reports and analyse fMRI scans. No Lie MRI is a proprietary product that

measures purpose, earlier knowledge, and dishonesty of brain activity by automatically analysing fMRI scans. The No Lie MRI studies the activity of central nervous system and produce the direct measure of truth verification and lie detection as in Figure 10.

Figure 10 Truth and lie detection using no lie MRI (see online version for colours)



2.4 Positron emission tomography (PET)

2.4.1 PET signal

Schmidt and Turkheimer (2002) discussed PET as a functional imaging scan which measures the functioning of organs, metabolic activity. It uses a radioactive drug called as TRACER which is introduced into the body to study the pharmacokinetics of radiotracers molecules. The tracers show up as a bright spot in PET scan in the area of high chemical activity. The CT scan can obtain 3D imaging of the tracer concentration. These radioisotopes will be injected into the blood stream after being attached to the compounds related to normal functioning of brain. The radioactive fluorine-18 (18F) and carbon-11(11C) are used to tag the sugar glucose. Oxygen-15 (15O) is used to tag water molecules and determine blood flow in brain. The designated detectors are placed around the patients head to collect the signals generated by radiotracers which are converted into 3D imaging of brain. PET is a non-invasive measuring procedure which enables the investigation of biochemical changes in the body of humans and animals. PET can be utilised to mark compounds of pharmacological and physiological importance as it can measure radioisotope concentrations in the nanomolar to picomolar range. These radiotracers can be utilised to understand neurochemical and metabolic procedures at the critical physiological concentrations.

Magistretti and Pellerin (1996) discussed diverse indications of brain activity, for example, cerebral blood flow (CBF), glucose use and oxygen utilisation are the prime signals monitored in brain imaging methods. These metabolic signs have been reviewed in different diseases, normally using (18F) fluoro-2 deoxyglucose (FDG) as a radiotracer in PET imaging. FDG is a widely used radiotracer for the past 32 years in PET studies.

The PET imaging FDG provisions non-invasive study of glucose consumption by various brain cells, such as neurons and glial cells.

2.4.2 Disease diagnose with PET

Fluorodeoxyglucose PET (FDG-PET) is used widely in early detection of AD and discriminating AD from various dementias. FDG-PET can provide deeper insight and track deterioration of AD; it can also recognise persons at danger for AD before the detection of cognitive symptoms.

Mosconi et al. (2010) discussed AD as perceived by an exact territorial pattern of combined metabolic rate of glucose (CMRGlc) decays. AD patients show constant CMRGlc shortfalls in the parieto-fleeting territories, posterior cingulate cortex (PCC) and medial temporal lobe (MTL). As the infection spreads, frontal affiliation cortices get to be included, but primary visual cortices, basal ganglia, cerebellum and striatum will not be affected. Knaapen et al. (2010) discussed cardiac PET is a non-invasive, nuclear imaging scan of the heart. It makes use radiotracers such as oxygen-15water, nitrogen-13 ammonia and rubidium-82 which yields images of the heart. Cardiac PET can identify coronary artery disease (CAD) and injuries caused due to cardiac arrest. PET scans differentiate between healthy and injured cardiac muscles. It is additionally used to discover percutaneous coronary intercession (PCI) like angioplasty and stenting, coronary artery bypass grafting (CABG).

Podoloff et al. (2007) discussed FDGPET as generally utilised in recognising and deciding diverse phases of numerous types of cancer, such as pancreatic, melanoma, colorectal, neck, head lung, esophageal, lymphoma, bosom, thyroid and cervical. In cancer patients there will be change in biochemistry of tissues and the cancer cells grow much rapidly than normal cells consuming the glucose. The patient who has cancer or suspected for the same is subjected to PET scan. The patient is injected with appropriate amount of FDG radiotracer composed of radioactive element and glucose. Once after the radiotracer is absorbed by the tissues underneath, the scan is conducted. The array of detectors in the scanner gather the signals generated by radiotracer using which the PET/CT scanner estimates the metabolic activity and represents it as the 3D imaging. This imaging helps doctors to understand the occurrence, aggressiveness of the disease and plan the treatment. PET/CT can determine the presence of active tumours, suspicious tissue masses and residual masses. Hence scanning is repeated after every cycle of the treatment to know the impact of the treatment and to make decisions about further treatments and surgery if needed.

2.4.3 PET devices in market

Siemens has developed Biograph mMR1 MR-PET. It combines the state-of-the-art 3T MRI with proven molecular imaging. It meets the requirements of PET in areas of Neurology, Oncology, Body Imaging, Cardiology and Pediatrics.

GE health care has developed SIGNA PET/MR. It comprises of 3.0T magnetic resonance technology and integrated PET technology. It can concurrently gather MR and PET data. MR is used for soft tissue imaging by capturing functional and morphological details. PET helps to study metabolism and cellular activity. The combination of MR and PET enables to diagnosis of cellular changes before any anatomical changes. Further the

anatomical images can be combined with biochemistry data to exactly determine the point of abnormal cell growth.

2.5 Near infra-red imaging

2.5.1 Near infra-red imaging signal

Jobsis (1977) discuss the fraction of electromagnetic spectrum, conscious aware by the human visual system is called the visible spectrum and its wavelength ranges from 380 to 700 nanometres. The near-infrared spectrum is situated as successor of red wavelength and consists wavelength that ranges from 700 to 1,100 nanometres and this depends on the application. There is no relationship between visible and NIR signal, even though the NIR band is located next to the visible one. It preserves high frequency information so that the human observer can easily interpret. Jobsis primarily illustrated that use of near infrared (NIR) light for non-invasively checking the state of cortical tissues through the skull. Following a decade, scientists recognised the utilisation of NIR spectroscopy for monitoring functional brain activity.

Jobsis (1977) discussed that in the observable and NIR wavelengths, oxygenated, deoxygenated hemoglobin, and water are the usual absorbents in the person's body. Hemoglobin strongly absorbs the visible wavelength of light, water absorption increases considerably near infrared above 950 nm and it decreases considerably at the wavelengths greater than 650 nm. Thus, the absorption of light is minor when the wavelength is in-between 650 and 950 nm and the light can scatter a few centimetres through the tissue, before it is distinguished. Jobsis described that the near infrared light can disperse all the way through the scalp and head of a mature human that can be used to find the hemoglobin concentrations in brain. Cope et al. (1988) used customised Beer-Lambert Law to register the absorption changes in the brain through absorption coefficient inside the purely absorbing samples and adding a differential path-length factor (DPF) to translate the enhanced path length of light through the highly distribution tissue. Huneau et al. (2015) discussed that a Neurovascular coupling is the correlation between variations in CBF due to local neural activity it depicts the modifications in blood flow and oxygen utilisation. Uludag et al. (2005) discussed that the raise in blood flow is directly proportional to increase in blood volume; hence more oxygenated hemoglobin is supplied to the downstream capillaries and veins to reduce deoxygenated hemoglobin. The increased oxygen utilisation leads to hike of deoxygenated hemoglobin concentration in turn reducing the oxygenated hemoglobin concentration. This association between the blood flow and oxygen utilisation during brain activity by and large results in an expansion of blood volume and oxygenated hemoglobin and in turn reduction of deoxygenated hemoglobin. Maki et al. (1995) measured the correlation between brain activity and the concentration of hemoglobin variation using near-infrared spectroscopy (NIRS). Maki et al. (1995) developed several instruments with set of sources and detectors for representing the hemoglobin variations related with brain activity.

The NIRS devices are inexpensive and is portable for functional neuroimaging compared to EEG, fMRI, PET and MEG. It is a non-invasive technique that uses laser light sources hence it is safer and does not require preventive measures. NIRS can determine the uncertainties in the blood flow against the oxygen consumption in response to brain activities by measuring the changes in oxygenated and deoxygenated

hemoglobin. NIRS can gauge hemodynamic signs with a temporal resolution of 100 Hz or more which is superior compared to fMRI or PET. This helps in filtering of physiological intervention from the required brain activity signals. NIRS is attuned with PET, EEG, MEG and fMRI enabling simultaneous multi-modular investigations of brain activities.

Here the author Naseer and Hong (2015) have concentrated on FNIRS-based BCI for signal generation, noise removal, feature extraction and classification technique.

2.5.2 Disease diagnose with near infra red

You et al. (2015) has discussed that NIRF as useful in diagnosis of lymphatic vascular diseases. The images of NIRF scan aids in recognising early tissue damages. The connected system of hollow cavities in the skull are called sinuses, usually they will be filled with air but during the infection it gets filled up with virus, fungus, liquids and microorganisms which is termed as Sinusitis. A digital imaging system is developed to study the health of maxillary sinuses by obtaining the image by the University of California, Irvine (UCI) researchers. The maxillary sinuses are illuminated uniformly by oral illuminator of NIR. The NIR light can easily penetrate through the clear sinuses. The presence of liquids in sinus reduces the intensity, and the path of the transmitted light will be recorded using NIR-sensitive digital cameras. Doctors can use this variable strength of light and spatial patterns to understand health of sinuses both in case of initial diagnosis and progress after treatment. This is first of its kind technique to diagnose sinusitis at low cost and avoid over-and under-treatments. Von Burstin et al. (2008) about pancreatic cancer is abnormal multiplication of the pancreatic cells, these cancerous cells may infect other parts of the body. It is the most dangerous disease as only 5% of the survivors lived for 5 years. This cancer do not show up early symptoms and there is lack of appropriate diagnostic gadgets for early discovery. Von Burstin et al. (2008) have utilised new atomic NIRF as a part of vivo imaging systems in clinically applicable mouse models of pancreatic cancer.

The group led by Iftimia et al. (2011) from Physical Sciences, Inc., has proved that NIRF imaging can reliably distinguish between low-risk and high-risk pancreatic cysts that may become malignant while other imaging techniques cannot classify the pancreatic cysts so accurately.

2.5.3 Near infra-red imaging devices in market

Though there are wide range of NIR devices available in the market from handheld to BED-based, we study the hand-held NIR devices because of their portability.

Tromberg et al. (1997) discussed that the first handheld NIR device which uses laser diodes (10–30 mw) is Tromberg. It is used to differentiate between the visual properties of usual and malignant injury containing breast tissue.

Liu et al. (2004) discussed PMT detector and a four light emitting diode-based NIR device which is used for wireless localisation of breast tumour.

Nioka and Chance (2005) as silicon diode detector which uses light emitting diodes in handheld NIR device. It is used to decide the sensitivity and specificity of breast cancer in human subjects.

Xu et al. (2007) developed a NIR device which uses laser diode and PMT for real-time monitoring reaction to dynamic compression stimuli and relate physiologic changes

2.6 Fetal Magnetoencephalography (fMEG)

2.6.1 fMEG signal

Blum et al. (1985) fMEG as a non-invasive method to determine brain signals of fetus in the womb. The fMEG records the magnetic field generated due to electric activity in the brain of the fetus using SQUIDs. The highly sensitive sensors capture brain activity signals of the fetus from maternal abdominal by filtering out the signals generated by Vernix Caseosa and other tissue around it.

The strength of the measured magnetic field depends on the distance between the source and the sensors. The noise due to fetal and maternal hearts is much intense compared to the magnetic field sensed by fMEG hence the interfering signals must be removed. Adaptive filtering and orthogonal signal projection algorithms are used to curb the noise. Once the cardiac noise is removed the remaining fMEG signal is analysed to find the physiologically meaningful patterns.

fMRI can also be used for examining fetus brain activity but with fMRI it is difficult to access as it involves safety issues.

2.6.2 fMEG devices

Philips has manufactured 31-channel-SQUID-biomagnetometer. Every channel comprises of a symmetrical gradiometer whose coil diameter is 20 mm and baseline is 70 mm. The gradiometer noise is around 10fT/Hz1/2at 1Hz.

2.7 Single photon emission computed tomography (SPECT)

2.7.1 SPECT signal

ALehyani (2009) discussed that SPECT is a medical imaging method; it is an amalgamation of nuclear medicine (NM) and CT. The SPECT scan needs gamma-emitting radionuclide to be injected to the patients. The gamma photons radiated from the radio-active source are perceived by radiation detectors (gamma cameras). The SPECT secures planar picture from various perspectives around the patient. The picture recreation strategies create cross segment pictures utilising the projection information. Compared to planer images of conventional NM the SPECT images deliver detailed information of the radionuclide distribution.

Saha (2016) discussed that SPECT uses a set of gamma cameras placed on a framework to provision the detector to record projections from various precise interims around the body. A coordinate system (r, s) is used to determine the extent of radioactivity at the point (x, y) that sums up to the signal recorded at location ‘r’ at rotation angle x in the projection acquired. The complete set of projection data is represented as two dimensional matrix p(r, x) using Sinogram where each row represents the strength of one projection and consecutive rows for successive projection angles.

The basic back-projection strategy is utilised to remake tomographic pictures which produces a 2D cross sectional picture by utilising the projection profiles from a part in the object.

Bousselham (2007) discussed that in the back-projection for each projection path the counts in each projection profile element is equally partitioned across the pixels. The summation of back projections for all profiles provides an estimate of the radioactivity dissemination within the scanned part. Filtered back-projection is adopted for reconstruction of tomographic images to obtain high quality images. Zhu et al. (1997) discussed that Fourier transform recreation is another technique for remaking of tomographic pictures where the amplitudes for various spatial frequencies are depicted in the FT of f(x) and the iterative reconstruction methods are applied.

2.7.2 SPECT disease diagnosis

Wang et al. (2012) highlighted that Parkinson's disease (PD) is a typical neurodegenerative illness which may bring about stiffness of the motor system, tremors, akinesia and changes in speaking and posture. The stiffness of the motor symptoms are because of the loss of dopaminergic neurons in the substantia nigra, pars compacta and the resulting inadequacy in striatal dopaminergic system.

SPECT neuroimaging can be used for diagnosing the early signs of the PD. This imaging can depict the disease extent, response to the treatment.

Sabharwal and Lahiri (2003) discussed SPECT can be used in myocardial perfusion imaging (MPI) of heart. It is a non-invasive diagnosis to depict the blood flow through the muscles of heart hence evaluating patients for CAD in situations such as checking myocardial feasibility before referring for myocardial re-vascularisation or patient with an irregular resting electrocardiogram (ECG) and confined activity resistance. SPECT has a proven success rate of 90% in detecting the CAD. This imaging technique has higher sensitivity detecting silent ischemia than stress ElectroCardioGram (ECG). Myocardial practicality can be evaluated by thallium re-infusion to give data in patient's extreme irreversible thallium imperfections on standard activity redistribution imaging.

Heiken (2007) discussed that hemangiomas are the most widely recognised benign lesion of the liver. There is a need for differentiating benign lesions from other malignant lesions. Chhieng (2004) discussed that percutaneous biopsy of liver hemangiomas is not used as there exists a risk of hemorrhage, so the non-invasive techniques are relied upon. Bonanno et al. (1991) discussed that SPECT can be used as a non-invasive diagnostic technique to differentiate hepatic lesions as hemangiomas. SPECT can easily differentiate between the blood pool of Hepatic hemangiomas and other solid hepatic masses using labelled red blood cells (RBCs) which are injected intravenously before scan.

Khan et al. (2004) have proposed that Carcinoids and other neuroendocrine tumours can be imaged with somatuline tagged with an appropriate radioisotope as they have somatostatin receptors. SPECT can be used for imaging (Octreoscan) of neuroendocrine tumours it is performed in conjunction with counter acting agent (antibody), generally at 24 years of age and once in a while at 48 hours after the infusion.

The SPECT can separate between territories of obsessive uptake and physiological uptake in the abdomen. SPECT can likewise separate amongst mesenteric and bone sores. The extra-planar pictures might be gotten from regions of particular enthusiasm, utilising longer exposure time for all the more effectively translated imaging.

Daffner et al. (2009) discussed that spondylolysis is an injuries brought because of anxiety breaks or push responses in the spine which can be imaged by bone filtering with SPECT. The SPECT is very effective in detecting early spondylolysis and anxiety fractures. It can recognise stress cracks days to weeks sooner than radiographs. Further can identify distinction amongst rigid and delicate tissue harm.

It is widely used in early detection of spondylolysis particularly in children and adolescents. SPECT imaging lack specificity, hence complementary imaging will be used for concluding the diagnosis. SPECT is used in differentiating between the lymphoma and death of cells in chest and abdomen. It is used in distinguishing cysts from other infectious tumours. SPECT can also differentiate between the swelling of bone and soft tissue's and hence is capable of detecting osteomyelitis.

2.7.3 SPECT devices

NeuroLogica has developed a portable SPECT device called inSPiraHD. The gadget can provide high resolution brain imaging. It can be used in different environments including ICU, OT, clinic and casuality, epilepsy; Parkinson's, stroke and Alzheimer's can be imaged using inSPira HD.

G-SPECT is a full-ring SPECT system developed by MILabs. It can provide high resolution imaging of brain, bone, paediatric and heart. It can work with different isotopes like 99 mTc, 111 In and 123I and it has < 3 mm isotropic resolution. Larger bore collimator option is provided in G-SPECT to obtain HD heart images.

Philips developed Bright View XCT which is a SPECT/CT system. It comprises of flat panel X-ray CT with co-planar design. It can produce high resolution imaging with low dose (1 mGy CTDIVOL) of radiotopes. The device can capture 15 different image sets concurrently and can capture heart volume without need for patient to hold the breath. The detector moves automatically near to the patients and the device is compact.

The Digirad Cardius 2 XPO SPECT cardiac imaging system. It comprises the dual-head cameras for faster imaging. The device uses high-definition solid-state detectors (HDSD) and innovative nuclear cameras to provide better separation between the heart and viscera. It is a compact and portable system.

3 Results and discussions

Figure 11 shows the temporal and spatial resolution of various non-invasive devices.

Table 4 and Table 5 include the comparisons of different non-invasive methods along with various parameters. Most of devices have been used in areas of diagnosis of neurological disorder, cardiac problem and spine related disorders.

Even though there are different neural modalities that provide information about the brain activity. The wave forms of brain activity are very important for the diagnosis, analysis and treatment of various diseases.

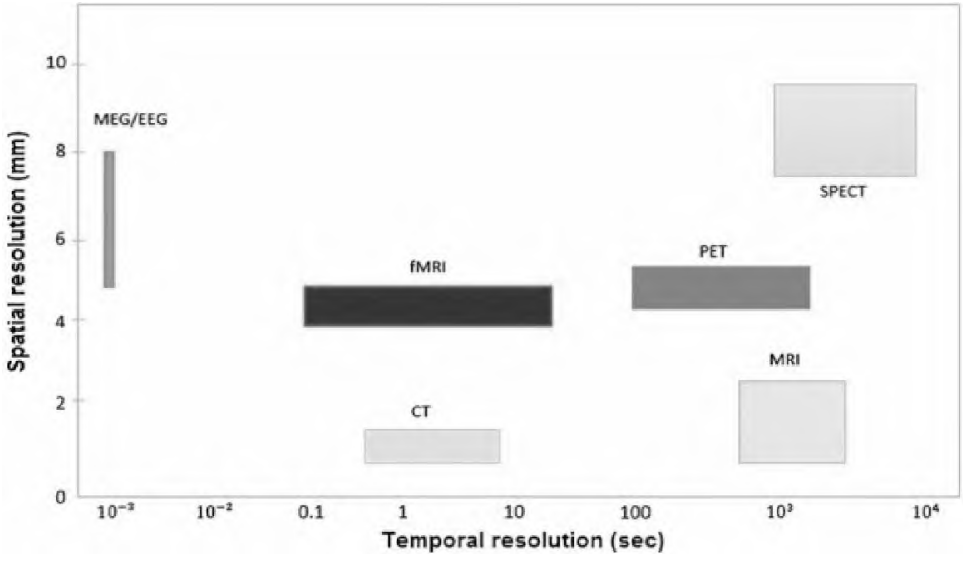
Table 4 Comparison of the different non-invasive acquisition techniques across the parameters, capability, cost, portability, safety and the available devices in market

Technology	*Capability*	*Device cost*	*Portability*	*Safety*	*Devices in the market*
EEG	Measures the brain activity at different instances due to the flow of electric current	As low as 100$ to 700$	Yes	Safe to wear and use	Emotiv, ULandT, SmartBrain, NeuroSky, Olimex EEGSMT,SIENNA ULTIMATE, Allengers VIRGO, EBNeuros Be Micro, MEDATECs Brainnet4, NIHON KOHDEN's Neurofax, NATUSs Schwarzer EEG, ATES Neurotrave, Neuro Prax EEG
MEG	Measure the brain activity of pyramidal cells of the sulci.	1 million $	No	Not safe for pregnant ladies and children	SQUID, VectorView, Elekta Neuromag
fMRI	Measures variations in, oxyhemoglobin and deoxyhemoglobin, ratio, measures raise in blood flow as a response to stimulus.	300,000$ to 1 million $	No	Implanted medical devices such as pacemakers could malfunction	Nata Technologies, NeuroDevice, BrainWave, No Lie, MRI, Cephos Corporation
NIRS	Measure activities near, surface of brain	1,000$	Yes	Safe to use	Tromberg, Liu, Chance
PET	Used to study the role of different neurotransmitters	1.8 million	No	Requires injection of radioactive tracers, so not suitable for everyone	Siemens Biograph, GE health cares SIGNA PET.
SPECT	High quality images with good contrast, provides depth information of radioisotope distribution compared to planar images	2 million	No	Requires injection of radioactive tracers, so not suitable for everyone	Requires injection of radioactive tracers, so not suitable for everyone

Table 5 Comparison of the different non-invasive acquisition techniques across the parameters activity measured, type of measurement, temporal and spatial resolution, advantages and drawbacks

Non-invasive acquisition methods	*Type of signal*	*Mode of acquisition*	*Temporal resolution in seconds*	*Spatial resolution in millimetres*	*Advantages*	*Disadvantages*
EEG	Electrical	Direct	~ 0.001 s	~ 0 mm	1 Simplicity of usage	1 Less signal to noise ratio and have poor spatial resolution
					2 Economical devices and have high temporal resolution	2 Does not provide the cross section of the brain
					3 Able to diagnose the diseases	3 Does not support brain imaging
					4 Harmless	
					5 Has restricted voltage for the device measuring	
MEG	Magnetic	Direct	~ 0.05 s	~ 5 mm	1 Higher spatial temporal, resolution, than EEG	1 The devices are structurally heavy and costly and simultaneous analysis is difficult with this devices
					2 Helps to identify the seizures and in patients with epilepsy	2 Expensive
					3 Magnetic fields are less influenced than electric signals	3 Sensitive to external noise
fMRI	Hemodynamic	Indirect	~1 s	~ 1 mm	1 These devices have high spatial resolution, no radiation exposure and are widely available	1 These devices provide Indirect measures of electrical activity in brain
					2 Inexpensive when compared to PET, EEG	2 They have poor temporal, resolution with a physiological delay of 3–6 sec
					3 Sensitive to motion	3 The head movement affect the signal quality
NIRS	Hemodynamic	Indirect	~ 1 s	~ 2 cm	1 Easily portable and economical devices	1 Least spatial resolution, delays introduced by hemodynamic response will reduce the communication speed.
					2 Moderate temporal, resolution, not require the use of electrodes and, conductive gel	2 Low temporal resolution
						3 Needs more time
SPECT	Hemodynamic	Indirect	~ 10–30 min	~ 1 cm	It has excellent spatial resolution and temporal resolution	SPECT lacks good resolution
PET	Hemodynamic	Indirect	~ 0.2 s	~ 1 mm	It has excellent spatial resolution	It has poor temporal resolution

Figure 11 Scale of temporal and spatial resolution of the different non-invasive acquisition techniques



Source: Hassanien et al. (2016)

Though there are different methods available for analysing brain functions like EEG, magnetoencephalography (MEG), fMRI, etc. EEG is suitable platform for monitoring the brain activities as it is cheap compared to other devices, easy to measure the EEG signals, less harmful compared to other modalities. EEG signals is also used to identify various diseases such as epilepsy, ALS, etc. EEG has also attracted medical health care centres to identify and diagnose of the diseases, entertainment to improve the concentration and meditation level of the people, criminal identification, academia and Industry. EEG has more advantages compared to other modalities, In future EEG can be used for diagnosing Physiological disease which helps the people in remote areas to cure there diseases at the earlier stage.

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References

Akareddy, S.M. and Kulkarni, P.K. (2013) 'EEG signal classification for epilepsy seizure detection using improved approximate entropy', *International Journal of Public Health Science (IJPHS)*, Vol. 2, No. 1, pp.23–32.

ALehyani, S. (2009) 'Application of single photon emission computed tomography (SPECT) parameters for bone scintigraphy', *Journal of King Saud University-Science*, Vol. 21, No. 2, pp.109–117.

Blum, T., Saling, E. and Bauer, R. (1985) 'First magnetoencephalographic recordings of the brain activity of a human fetus', *BJOG: An International Journal of Obstetrics & Gynaecology*, Vol. 92, No. 12, pp.1224–1229.

Bonanno, N., Baldari, S., Cerrito, A., Zimbaro, G., Restifo, G., Blandino, A. and Freni, O. (1991) 'Diagnosis of hepatic hemangiomas with 99mtc-labeled red blood cell scanning: value of Spect', *Journal of Nuclear Biology and Medicine* (1991, Turin, Italy), Vol. 35, No. 3, pp.135–140.

Boukari, N. and Djemili, R. (2016) 'Combined strategy neural network using AR parameters for Epileptic EEG signals classification', *International Journal of Biomedical Engineering and Technology*, Vol. 21, No. 1, pp.67–79.

Bousselham, A.K. (2007) *FPGA Based data Acquisition and Digital Pulse Processing for PET and SPECT*', PhD dissertation, Fysikum.

Chhieng, D.C. (2004) 'Fine needle aspiration biopsy of liver-an update', *World Journal of Surgical Oncology*, Vol. 2, No. 1, p.5.

Chowdhury, R.A., Lina, J.M., Kobayashi, E. and Grova, C. (2013) 'MEG source localization of spatially extended generators of epileptic activity: comparing entropic and hierarchical Bayesian approaches', *PloS One*, Vol. 8, No. 2.

Chu, R.K., Braun, A.R. and Meltzer, J.A. (2015) 'Meg-based detection and localization of perilesional dysfunction in chronic stroke', *NeuroImage: Clinical*, Vol. 8, pp.157–169.

Cohen, D. (1968) 'Magnetoencephalography: evidence of magnetic fields produced by alpha-rhythm currents', *Science*, Vol. 161, No. 3843, pp.784–786.

Cohen, D. (1972) 'Magnetoencephalography: detection of the brains electrical activity with a superconducting magnetometer', *Science*, Vol. 175, No. 4022, pp.664–666.

Cope, M., Delpy, D.T., Reynolds, E.O.R., Wray, S., Wyatt, J. and Van der Zee, P. (1988) 'Methods of quantitating cerebral near infrared spectroscopy data', in *Oxygen Transport to Tissue X*, pp.183–189, Springer, Boston, MA.

Daffner, R.H., Weissman, B.N., Bennett, D.L., Blebea, J.S. and Jacobson, J.A. (2009) 'Expert panel on musculoskeletal imaging, American college of radiology appropriateness criteria: stress/insufficiency fracture, including sacrum, excluding other vertebrae'.

Graimann, B., Allison, B and Pfurtscheller, G. (2009) 'Brain-computer interfaces: a gentle introduction', in *Brain-Computer Interfaces*, pp.1–27, Springer, Berlin, Heidelberg.

Hassanien, A.E., Shaalan, K., Gaber, T., Azar, A.T. and Tolba, F. (2016) *Proceedings of the International Conference on Advanced Intelligent Systems and Informatics 2016*, Springer, Vol. 533.

Hatsusaka, N., Higuchi, M. and Kado, H. (2009) 'Possibility of MEG as an early diagnosis tool for Alzheimer's disease: a study of event related field in missing stimulus paradigm', in *13th International Conference on Biomedical Engineering*, Springer, pp.9–12.

Heiken, J.P. (2007) 'Distinguishing benign from malignant liver tumours', *Cancer Imaging*, Vol. 7, No. Spec No A, pp.S1–14.

Huneau, C., Benali, H. and Chabriat, H. (2015) 'Investigating human neurovascular coupling using functional neuroimaging: a critical review of dynamic models', *Frontiers in Neuroscience*, Vol. 9, No. 9, p.467.

Iftimia, N., Cizginer, S., Deshpande, V., Pitman, M., Tatli, S., Iftimia, N.A. and Brugge, W.R. (2011) 'Differentiation of pancreatic cysts with optical coherence tomography (oct) imaging: an ex vivo pilot study', *Biomedical Optics Express*, Vol. 2, No. 8, pp.2372–2382.

James, L.M., Belitskaya-L'evy, I., Lu, Y., Wang, H., Engdahl, B.E., Leuthold, A.C. and Georgopoulos, A.P. (2015) 'Development and application of a diagnostic algorithm for posttraumatic stress disorder', *Psychiatry Research: Neuroimaging*, Vol. 231, No. 1, pp.1–7.

Jobsis, F.F. (1977) 'Noninvasive, infrared monitoring of cerebral and myocardial oxygen sufficiency and circulatory parameters', *Science*, Vol. 198, No. 4323, pp.1264–1267.

Khan, A., Jones, C. and MacDonald, S. (2004) *Asbestos-Related Disease* [online] http://www.emedicine.com/radio/topic53.htm.

Kidwai, M.S. and Saeed, S.H. (2017) 'A novel approach to study the effects of anesthesia on respiratory signals by using the EEG signals', *International Journal of Electrical and Computer Engineering (IJECE)*, Vol. 7, No. 6, pp.3313–3317.

Knaapen, P., De Haan, S., Hoekstra, O.S., Halbmeijer, R., Appelman, Y.E., Groothuis, J.G.J. and Gotte, M.J.W. (2010) 'Cardiac pet-ct: advanced hybrid imaging for the detection of coronary artery disease', *Netherlands Heart Journal*, Vol. 18, No. 2, pp.90–98.

Larsen, E.A. (2011) 'Classification of EEG signals in a brain-computer interface system', Master's thesis, Institutt for datateknikk og informasjonsvitenskap.

Liu, Q., Luo, Q. and Chance, B. (2004) '2D phased array fluorescence wireless localizer in breastcancer detection', in *2003 IEEE International Workshop on Computer Architectures for Machine Perception, IEEE*, pp.71–73.

Logothetis, N.K. (2002) 'The neural basis of the blood-oxygen-level-dependent functional magnetic resonance imaging signal', *Philosophical Transactions of the Royal Society of London B: Biological Sciences*, Vol. 357, No. 1424, pp.1003–1037.

Magistretti, P. and Pellerin, L. (1996) 'The contribution of astrocytes to the 18f-2-deoxyglucose signal in pet activation studies', *Molecular Psychiatry*, Vol. 1, No. 6, pp.445–452.

Maki, A., Yamashita, Y., Ito, Y., Watanabe, E., Mayanagi, Y. and Koizumi, H. (1995) 'Spatialand temporal analysis of human motor activity using noninvasive NIR topography', *Medical Physics*, Vol. 22, No. 12, pp.1997–2005.

Mosconi, L., Berti, V., Glodzik, L. Pupi, A., De Santi, S. and de Leon, M.J. (2010) 'Pre-clinical detection of Alzheimer's disease using fdg-pet, with or without amyloid imaging', *Journal of Alzheimer's Disease*, Vol. 20, No. 3, pp.843–854.

Naseer, N. and Hong, K.S. (2015) 'fNIRS-based brain-computer interfaces: a review', *Frontiers in Human Neuroscience*, Vol. 9, p.3.

Nicolas-Alonso, L.F. and Gomez-Gil, J. (2012) 'Brain computer interfaces, a review', *Sensors*, Vol. 12, No. 2, pp.1211–1279.

Nioka, S. and Chance, B. (2005) 'Nir spectroscopic detection of breast cancer', *Technology in Cancer Research & Treatment*, Vol. 4, No. 5, pp.497–512.

Ogawa, S., Menon, R., Tank, D., Kim, S., Merkle, H., Ellermann, J. and Ugurbil, K. (1993) 'Functional brain mapping by blood oxygenation level-dependent contrast magnetic resonance imaging: a comparison of signal characteristics with a biophysical model', *Biophysical Journal*, Vol. 64, No. 3, pp.803–812.

Palaniappan, R. (2014) 'Electroencephalogram-based brain-computer interface: an introduction', in *Guide to Brain-Computer Music Interfacing*, pp.29–41, Springer, London.

Pedreira, C., Martinez, J and Quiroga, R.Q. (2009) 'Neural prostheses: linking brain signals to rosthetic devices', *Proceedings on the ICROS-SICE International Joint Conference*, Ukuoka, Japan.

Podoloff, D.A., Advani, R.H., Allred, C., Benson, A.B., Brown, E., Burstein, H.J. and Edge, S.B. (2007) 'NCCN task force report: positron emission tomography (pet)/computed tomography (CT) scanning in cancer', *J. Natl. Compr. Canc. Netw.*, Vol. 5, No. 1, pp.S1–22.

Pritika and Budhiraja, S. (2016) 'Multimodal medical image fusion based on guided filtered multi-scale decomposition', *International Journal of Biomedical Engineering and Technology*, Vol. 20, No. 4, pp.285–301.

Ramadan, R.A. and Vasilakos, A.V. (2017) 'Brain computer interface: control signals review', *Neurocomputing*, Vol. 223, pp.26–44.

Sabeti, M., Behroozi, R and Moradi, E. (2016) 'Analysing complexity, variability and spectral measures of schizophrenic EEG signal', *International Journal of Biomedical Engineering and Technology*, Vol. 21, No. 2, pp.109–127.

Sabharwal, N. and Lahiri, A. (2003) 'Role of myocardial perfusion imaging for risk stratification in suspected or known coronary artery disease', *Heart*, Vol. 89, No. 11, pp.1291–1297.

Saha, G.B. (2016) 'Cyclotron and production of pet radionuclides', in *Basics of PET Imaging*, pp.143–160, Springer.

Sato, J.R. Basilio, R., Paiva, F.F., Garrido, G.J., Bramati, I.E., Bado, P. and Moll, J. (2013) 'Real-time fMRI pattern decoding and neurofeedback using FRIEND: an FSL – integrated BCI toolbox', *PLoS One*, Vol. 8, No. 12, p.e81658.

Schmidt, K and Turkheimer, F. (2002) 'Kinetic modeling in positron emission tomography', *The Quarterly Journal of Nuclear Medicine and Molecular Imaging*, Vol. 46, No. 1, p.70.

Shellock, F.G. and Kanal, E. (1996) *Magnetic Resonance: Bioeffects, Safety, and Patient Management*, Lippincott Williams & Wilkins.

Sheridan, C.J., Matuz, T., Draganova, R., Eswaran, H. and Preissl, H. (2010) 'Fetal magnetoencephalography achievements and challenges in the study of prenataland early postnatal brain responses: a review', *Infant and Child Development*, Vol. 19, No. 1, pp.80–93.

Singh, S and Bansal, D. (2016) 'Development of ocular motory movements-based online BCI model with multi-triggering capabilities for rehabilitative control', *International Journal of Biomedical Engineering and Technology*, Vol. 21, No. 3, pp.295–310.

Singla, R and Jatana, S. (2017) 'The types of hybrid modalities in brain-computer interface systems: a review', *International Journal of Biomedical Engineering and Technology*, Vol. 23, No. 1, pp.48–58.

Smith, S. (2005) 'EEG in neurological conditions other than epilepsy: when does it help, what does it add', *Journal of Neurology, Neurosurgery & Psychiatry*, Vol. 76, No. 2, pp.ii8–ii12.

Tromberg, B.J., Coquoz, O., Fishkin, J.B., Pham, T., Anderson, E.R., Butler, J. and Pham, D. (1997) 'Non-invasive measurements of breast tissue optical properties using frequency-domain photon migration', *Philosophical Transactions of the Royal Society of London B: Biological Sciences*, Vol. 352, No. 1354, pp.661–668.

Uludag, K., Dubowitz, D. J. and Buxton, R. B. (2005) 'Basic principles of functional MRI', *Clinical MRI*, pp.249–287, Elsevier, San Diego.

Von Burstin, J., Eser, S., Seidler, B., Meining, A., Bajbouj, M., Mages, J. and Schneider, G. (2008) 'Highly sensitive detection of early-stage pancreatic cancer by multimodal near-infrared molecular imaging in living mic', *International journal of cancer*, Vol. 123, No. 9, pp.2138–2147.

Wang, L., Zhang, Q., Li, H. and Zhang, H. (2012) 'Spect molecular imaging in Parkinson's disease', *BioMed Research International*.

Xu, R.X., Qiang, B., Mao, J.J. and Povoski, S.P. (2007) 'Development of a handheld near-infrared imager for dynamic characterization of in vivo biological tissue systems', *Applied Optics*, Vol. 46, No. 30, pp.7442–7451.

You, J.S., Cerussi, A E., Kim, J., Ison, S., Wong, B., Cui, H. and Bhandarkar, N. (2015) ' Near-infrared imaging for management of chronic maxillary sinusitis', in *SPIE BiOS*, International Society for Optics and Photonics, pp.93 140D–93 140D.

Zhu, J., Penczek, P.A., Schroder, R. and Frank, J. (1997) 'Three-dimensional reconstruction with contrast transfer function correction from energy-filtered cryoelectron micrographs: procedure and application to the 70sescherichia coliribosome', *Journal of Structural Biology*, Vol. 118, No. 3, pp.197–219.

Zimmerman, J.E., Thiene, P. and Harding, J.T. (1970) 'Design and operation of stable RF-biased superconducting point-contact quantum devices, and a note on the properties of perfectly clean metal contacts', *Journal of Applied Physics*, Vol. 41, No. 4, pp.1572–1580.